

Assignment 3

Copyright reserved

Electricity and Electronics EBN 122

18 August 2023

Test Information

Maximum marks:	20	Full marks:	20
Due Date:	25 August, 23:59	Open/Closed book:	Open
Total number of pages (this page included)		11	

Lecturers: Prof P. de Villiers, Prof. P.J Jacobs

Assistant Lecturers: N.-J Lord, M. Malesela, G. Neeuwfan

Important

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your assessment, you declare the following:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

I hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

6. Contact ebn2023.consult@gmail.com **before** the deadline for assistance with technical difficulties.

Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \deg$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

Section 1

Question 1 [2]

Given: k

$\begin{bmatrix} 5 & -5 & 0 \\ -2 & 20 & 0 \\ 0 & 0 & k \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ -10 \end{bmatrix}$
--

01	Calculate i_3 [2]
-----------	---------------------

Question 2 [2]	
Given: v_1	
$\begin{bmatrix} 6 & 3 & -10 \\ -3 & 0 & \beta \\ 2 & -8 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 2 \end{bmatrix}$	
02	Calculate β [2]

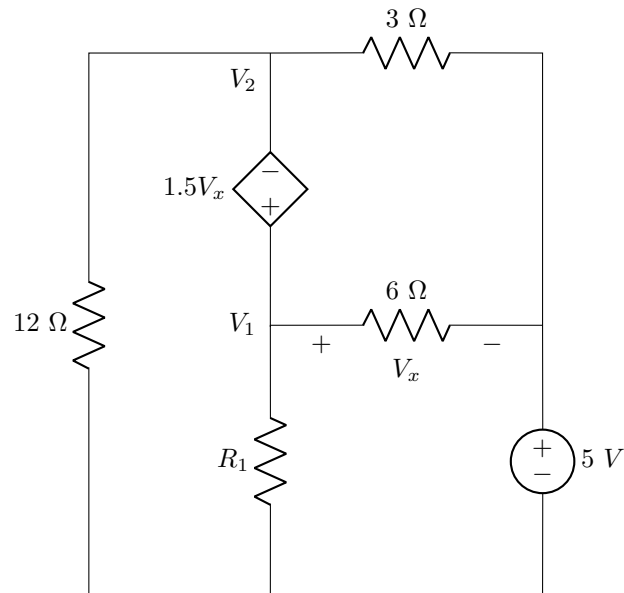
Question 3 [1]	
Given: k	
$\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} k \\ -5 \end{bmatrix}$	
03	Calculate V_1 [1]

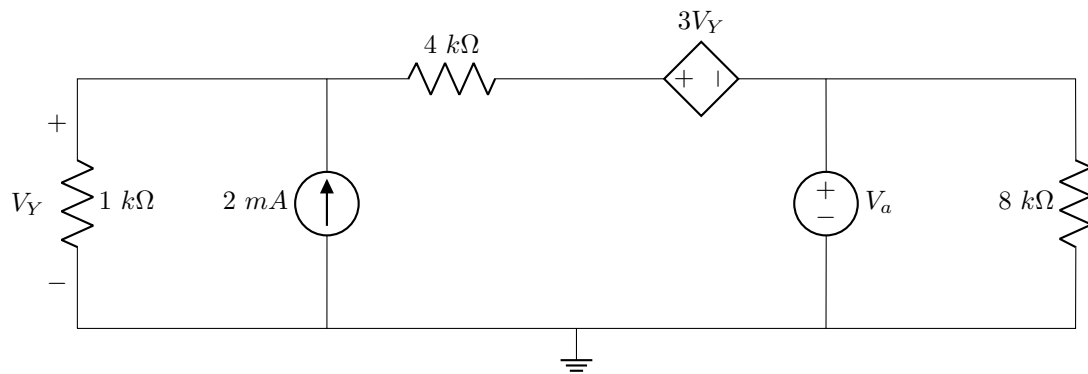
Question 4 [1]	
Given: a	
$\begin{bmatrix} k & -3 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} a \\ 10 \end{bmatrix}$	
04	Calculate k if $I_1 = 1 \text{ A}$ [1]

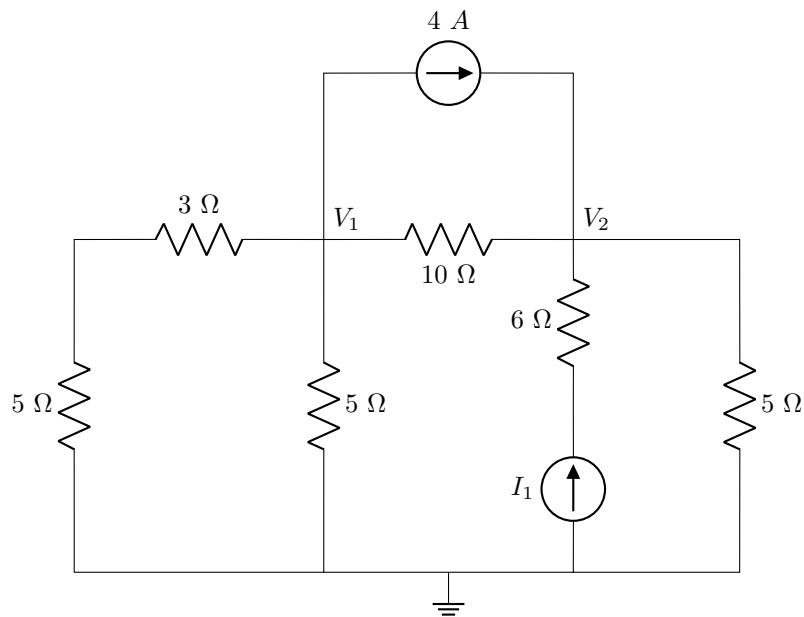
Questions 5 to 8 [4]Given: A , R_1 .Use nodal analysis to set up two equations of the following form and find co-ordinates a , b , c and d .

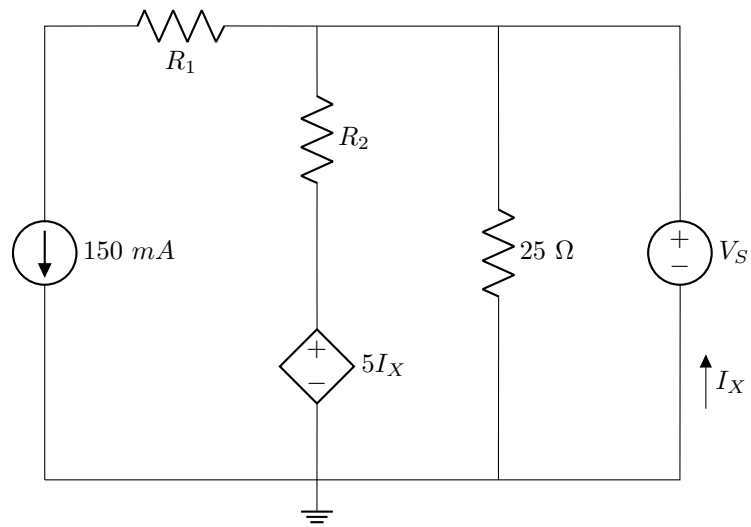
$$aV_1 + bV_2 = A$$

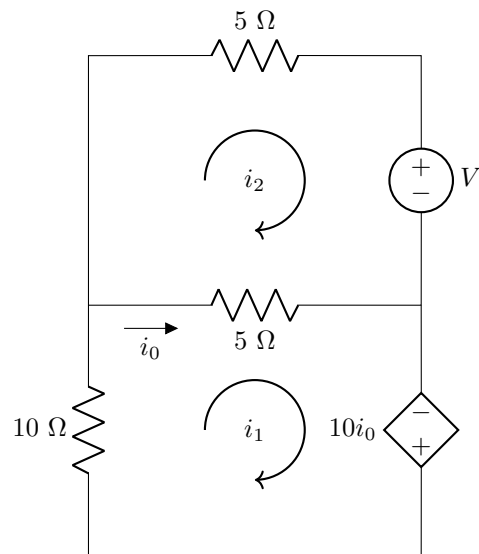
$$cV_1 + dV_2 = -7.5$$

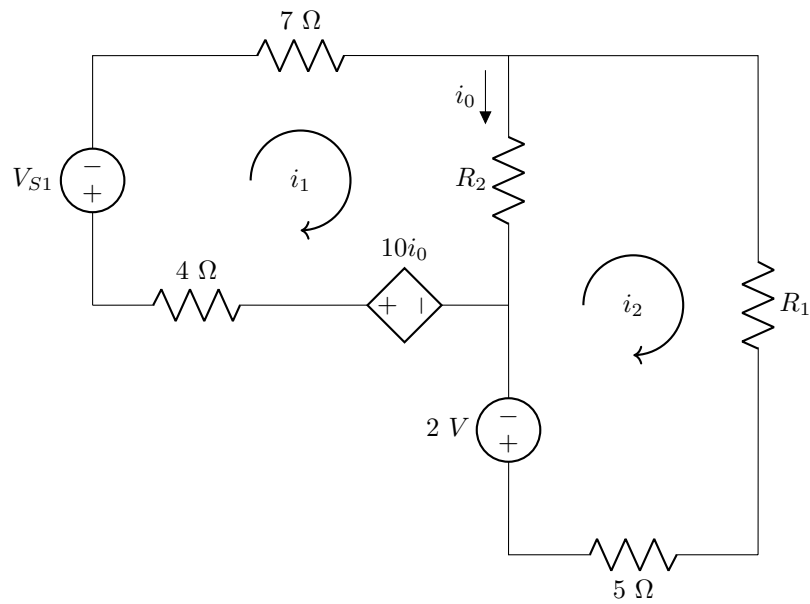
**05** a [1]**06** b [1]**07** c [1]**08** d [1]

Question 9 [2]Given: V_a **09** Use nodal analysis to determine V_Y [2]

Question 10 [2]Given: I_1 .**10** Use nodal analysis to determine V_1 [2]

Question 11 [2]Given: R_1 , R_2 , V_S .**11** Use nodal analysis to find I_X [2]

Question 12 [2]Given: V .**12** Using mesh analysis to find i_1 [2]

Question 13 [2]Given: V_{S1} , R_1 , R_2 .**13** Determine i_0 [2]**Total Section 1: [20]****Grand Total : [20]**